



Background-Aware Sensitivity Analysis for Heavy Neutral Leptons at ProtoDUNE-HD

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Abstract

Heavy neutral leptons (HNLs) provide a well-motivated extension of the Standard Model and offer a possible explanation for the small but non-zero masses of active neutrinos [2, 3]. The CERN Super Proton Synchrotron and the downstream ProtoDUNE-HD detector provide a potential configuration for producing HNLs in meson decays and observing their subsequent decays inside the detector [5, 10]. In this work we present a background-aware, truth-level analysis of the corresponding HNL sensitivity. HNLux signal samples are studied for electron-dominant and muon-dominant mixing, while beam-induced neutrino interactions simulated using GENIE provide the Standard Model background [8, 6]. We divide events into three two-charged-particle topologies for our analysis, and apply kinematic cuts to all events, providing discriminating power to reject up to 99.9% of background at certain masses. These results establish a background-aware benchmark for HNL searches at ProtoDUNE. The strongest projected sensitivities reach $|U_{eN}|^2 = 7.0 \times 10^{-8}$ at $m_N = 0.406$ GeV for electron mixing and $|U_{\mu N}|^2 = 1.8 \times 10^{-7}$ at $m_N = 0.349$ GeV for muon mixing over the corresponding accessible mass ranges.

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1 Introduction

The evidence for neutrino oscillations reported in 1998 by Super-Kamiokande established that at least two neutrino mass eigenstates are non-degenerate, and that at least one has a small but non-zero mass [1]. This was direct evidence for the requirement of an extension beyond the minimal Standard Model (SM), in which neutrinos are massless [2]. One possible extension is to introduce gauge-singlet neutral fermions which permit both a neutrino Yukawa interaction and a gauge-invariant Majorana mass. Through the type-1 seesaw neutrino mass mechanism, light neutrino masses are generated, along with singlet heavier-mass eigenstates known as Heavy Neutral Leptons (HNL) [2, 3]. As a result of their mixing with active neutrino flavour states, searches for HNLs could reveal an entirely new set of particles and provide a way to test fundamental physics connected to the origin of neutrino masses.

Crucially, HNLs can be produced and decay by weak interactions, while being long-lived enough to travel long distances before decaying [4]. A 2024 signal-only sensitivity study identified ProtoDUNE as a potential facility for HNL searches, focusing its numerical analysis on NP02 [5]. Here we study the horizontal-drift ProtoDUNE detector at CERN (ProtoDUNE-HD, CERN experiment NP04) [12]. HNLs produced in meson decays following SPS proton collisions could reach the detector, whose large liquid-argon active volume offers a region in which they may decay visibly [5].

The central question in this report is whether hypothesised HNL decays inside ProtoDUNE-HD can be distinguished from neutrino interaction background signals producing the same visible two-particle topologies. To answer this, we present a background-aware, truth-level projection of the sensitivity of ProtoDUNE-HD to HNLs. Building on the 2024 feasibility study [5], we use GENIE [6] simulations of beam-neutrino interactions to model background samples, and include topological selections, kinematic cuts, and background-inclusive sensitivities. We use supplied HNLux signal samples for HNL masses 0.010–1.967 GeV (up to 1.862 GeV in the muon-mixing samples), across different meson production channels, for both muon and electron dominant mixing hypotheses [8]. The mass grids and signal-file selections used for the three background-aware topologies are recorded in the analysis manifests. Although we work with truth-level data, throughout the analysis we consider each HNL decay channel using detector-motivated requirements, including grouping channels by visible two-particle topologies and implementing an analysis-defined charged-particle energy threshold. Ideally this provides a basis for a later sensitivity study, such that our work can be adapted for the full experimental sensitivity projection, including detector response and full particle reconstruction.

It must be re-emphasised that detector response, event reconstruction, particle misidentification, trigger and timing efficiencies, and cosmic-ray backgrounds are not simulated. On top of this, the beam-neutrino uncertainty is only represented through a simplified normalisation treatment. The resulting work should therefore be interpreted in this context as a background-aware truth-level benchmark rather than a complete experimental sensitivity. In the next section we describe the layout of the experimental facility, the detection process in the NP04 liquid argon detector, and the expected neutrino and HNL interaction modes within our detector. In the sections following that, we then present our analysis channels, kinematic cuts procedure, and finally the full sensitivity results.

2 The SPS–ProtoDUNE Search Environment

The Super Proton Synchrotron (SPS) supplies both the Large Hadron Collider (LHC) and CERN's North Area, but the protons relevant here are extracted to the North Area, where 400 GeV protons strike the T2 target, upstream of ProtoDUNE-HD (NP04) [9, 5]. Collisions at T2 produce secondary

mesons, including charged pions, kaons, D and D_s mesons, whose decays could produce an HNL instead of an ordinary neutrino if the HNL mixes with an active neutrino [5, 7]. An HNL emitted towards NP04 travels roughly 723 m from T2 to the detector, passing through shielding and soil without leaving a track [5], meaning that only those with a suitable lifetime and direction could reach the detector and decay inside it. The parent meson and the other particles in its decay also determine which HNL masses can be produced [7, 3]. A schematic of the overall arrangement of the experimental area is shown in Figure 1.

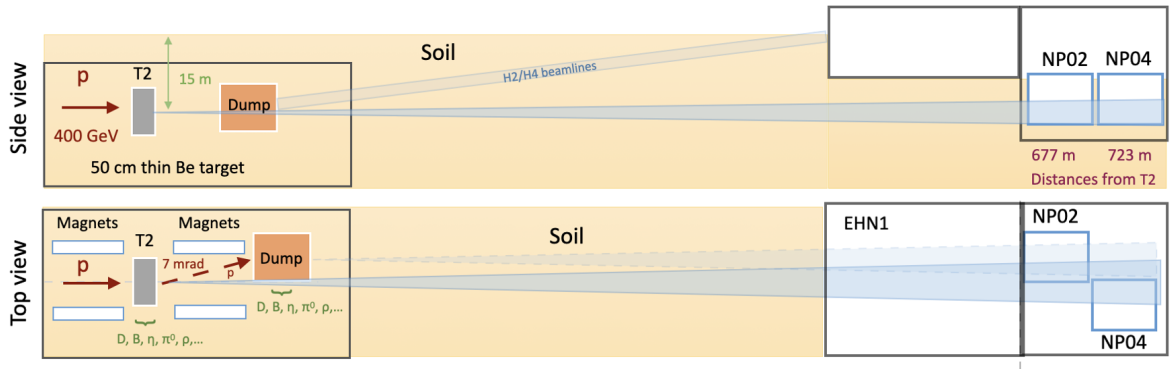


Figure 1: Schematic side and top views of the SPS–ProtoDUNE layout (not to scale). The T2 target, downstream TAX dump, intervening soil and NP04 in EHN1 are shown; NP04 is labelled 723 m from T2. Reproduced from Fig. 1 of Coloma *et al.*, JHEP 01 (2024) 134 [5]. The figure also appears in A. Chatterjee, "BSM searches @ProtoDUNE-SPS," LLP25 (2025), slide 6 [10].

The NP04 detector is a horizontal-drift liquid argon time projection chamber [12]. The active argon volume allows particles to interact and records the tracks of charged particles that travel through it. It is able to do this as charged particles crossing the argon leave ionisation electrons and produce scintillation light. An electric field then moves the electrons horizontally towards wire readout planes on either side of a central cathode; signals from the differently oriented wires, together with the drift time, allow the path and energy deposit to be reconstructed in three dimensions [11, 12]. Photon detectors record the emitted light and provide timing information for events [11, 12]. An HNL would not leave a track characteristic of a charged particle, but a decay inside the active volume could appear as charged particles emerging from a common vertex [4]. These detector capabilities motivate us to look for visible charged daughters and to use their multiplicity and kinematics; the analysis that we develop in the following section applies those ideas at truth level rather than simulating the readout or reconstruction.

The HNL decays of interest have the forms $N \to \ell^{\pm}\pi^{\mp}$ and $N \to \nu\ell_{\alpha}^{\pm}\ell_{\beta}^{\mp}$, where the charged leptons are electrons or muons [3, 7]. In either case, NP04 could image the two primary charged daughters as tracks or showers [11, 12], while the light neutrino in the second decay form would pass through essentially unseen.

The SPS collisions at T2 also produce mesons that decay to ordinary neutrinos, some of which travel towards NP04 and interact with its argon [10]. These beam-induced neutrino-argon interactions are the Standard Model background that we include in our HNL sensitivity calculation. We model them with four wNP04 GENIE background files supplied for this analysis, one for each incident neutrino type: ν_e, $\bar{\nu}_e$, ν_μ and $\bar{\nu}_\mu$ [6, 10]. Charged-current interactions can produce a charged lepton and hadrons, while neutral-current interactions can produce hadrons without an outgoing charged lepton [6]. Certain neutrino-argon background interactions can mimic the decay signals that we would expect from HNL decays in the argon active volume. As such, with the goal of eliminating false positives from background interactions, the main section of our analysis uses these four files to select signal-like background events,

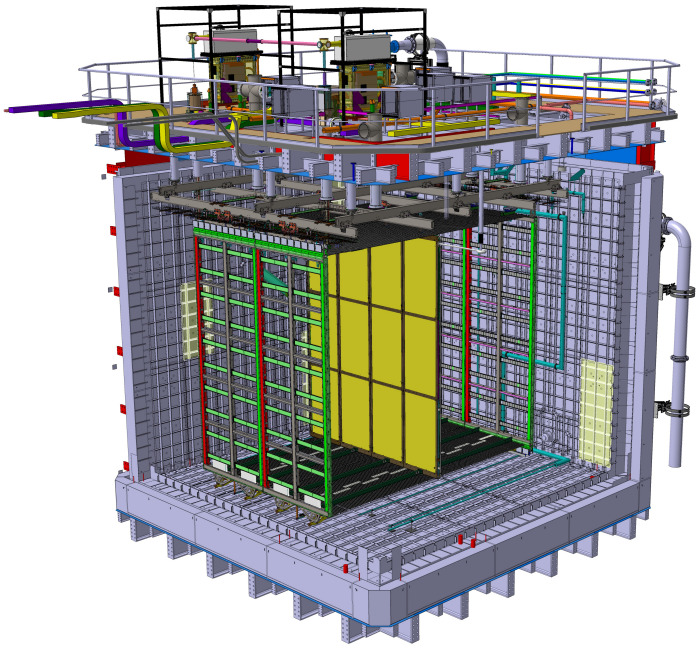


Figure 2: Schematic of ProtoDUNE-HD showing the central cathode (yellow) between the liquid-argon drift volumes and the surrounding wire anode plane assemblies. Ionisation electrons drift horizontally through each volume towards the anodes. Cropped from Fig. 20 of DUNE Collaboration, JINST 21 (2026) P08018 [12].

weight their expected yields and determine how many survive the kinematic cuts before calculating the background-inclusive sensitivity. Because NP04 is at the surface, cosmic rays are another potential source of tracks [5, 10]. We omit cosmic-ray backgrounds in this truth-level benchmark and assume they can be sufficiently suppressed; this assumption requires a dedicated study. Section 3 defines the common signal and background topologies by which we consider our data and the selections that we apply to them.

3 Analysis Framework

The aim of our analysis is to identify GENIE neutrino backgrounds that mimic HNL signal topologies, suppress them using kinematic selections, and calculate the background-inclusive sensitivity to $|U_{\alpha N}|^2$. Overall, the analysis can be broken down into the following steps in order to achieve this. Firstly, in Section 3.2, select events from the neutrino-argon background interaction data that mimic HNL signal event topologies. Secondly in Section 3.3, apply cuts to the data based on kinematic variables that provide distinguishing power between signal and background. Finally in Section 3.4, calculate expected background and signal yields for the events that pass the kinematic cuts, and use these to calculate the sensitivity curve.

3.1 Visible Charged-Particle Topologies

In the following table we define the visible charged-particle topologies that we need before we can proceed with the analysis. We also specify the HNL decay channels represented in the simulated samples, grouped by their two particle topologies [7, 8].

Following the muon-like topology convention used in an ICARUS-like HNL study [4], we count either a charged muon or a charged pion as one "muon-like" particle, since we have not simulated the track reconstruction and particle identification needed to separate them. Thus 2μ-like includes $\mu\mu$, $\mu\pi$ and $\pi\pi$, while 1μ-like $+1e$ includes $e\mu$ and $e\pi$.

Topology	**HNL signal decay channels**	**HNL parent mesons**	**GENIE charged pairs**
2μ-like	$N \to \nu\mu^+\mu^-$ (U_e, U_μ); $N \to \mu^\pm\pi^\mp$ (U_μ)	$K^\pm$, $D^\pm$, $D_s^\pm$	$\mu\mu$, $\mu\pi$, $\pi\pi$
1μ-like $+1e$	$N \to \nu e^\pm\mu^\mp$ (U_e, U_μ); $N \to e^\pm\pi^\mp$ (U_e)	$\pi^\pm$, $K^\pm$, $D^\pm$, $D_s^\pm$	$e\mu$, $e\pi$
$2e$	$N \to \nu e^+e^-$ (U_e, U_μ)	$\pi^\pm$, $K^\pm$, $D^\pm$, $D_s^\pm$	ee

Table 1: The three visible charged-particle topologies used for the background-aware HNL signal and GENIE neutrino-argon analysis. U_e and U_μ identify the electron- and muon-mixing signal samples in which each decay channel is represented. In the third column we list the meson families present in at least one signal channel for that topology; their availability depends on the decay channel, mixing hypothesis and HNL mass. The separate all-channel signal-only reference sensitivities in Figures 5–7 also include $N \to \nu\pi^0$ and $N \to \ell^\pm\rho^\mp$ modes, which are outside these three background-aware topologies.

These two-particle topologies are motivated by detector and analysis constraints. Daughter neutrinos, for example, where present, are effectively invisible, so we classify the decays by their two primary charged particles [3].

3.2 Event Selection

Our first selection stage identifies GENIE neutrino interactions with the same low-multiplicity charged-particle signature as the HNL decays in Table 1. It is applied event by event to the final-state GENIE record, before any pair-kinematic cuts. Figure 3 summarises the procedure used to make the background samples for the final sensitivity plots.

For each final-state particle, the kinetic energy is $K = E - m$, using its PDG mass for ordinary charged particles [13] and the mass inferred from its four-vector for a charged nuclear remnant. The analysis-defined 40 MeV generator-level threshold is applied to all charged species present in the simulated samples. Thus an extra detectable proton or other charged hadron causes an otherwise suitable event to fail the exact-two requirement. Neutral particles, including neutrons, photons and neutrinos, are not counted and are not separately vetoed. Photons from neutral-pion decays can convert into charged pairs in liquid argon [4]; this detector effect is not modelled here.

The two counted charged particles must match one of the categories in Table 1: ee for $2e$; $e\mu$ or $e\pi$ for $1e + 1\mu$-like (`1m`); and $\mu\mu$, $\mu\pi$ or $\pi\pi$ for 2μ-like (`2m`). Events with exactly two detectable charged particles in any other pairing are outside these analysis regions. Across the four GENIE flavour samples, 62,891 of the 800,000 generated interactions (7.86%) enter one of the three categories.

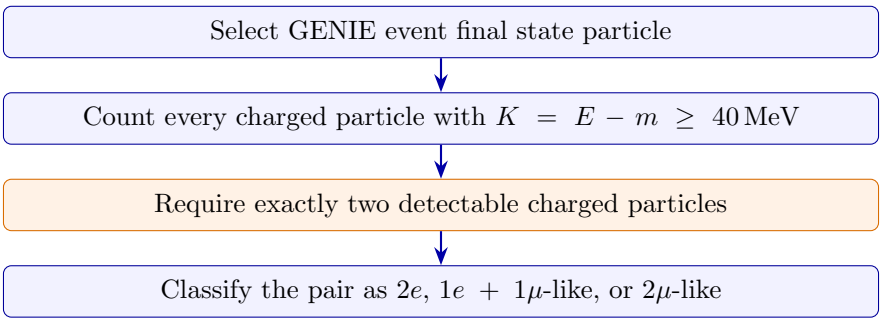


Figure 3: The GENIE background selection process by topology. Neutral particles and charged particles below energy threshold do not enter the charged multiplicity. The selected pair is subsequently subjected to the mass-dependent kinematic cuts of Section 3.3.

3.3 Kinematic Cuts and Background Rejection

For boosted, forward-going HNLs, visible daughters tend to be more collimated and to carry less combined transverse momentum than a topology-matched neutrino interaction, motivating these variables as in an ICARUS-like HNL study [4]. In our simulated samples, the pair-energy distribution also changes with HNL mass and parent meson. We can take advantage of these differences, motivating us to cut events using upper thresholds on pair energy, transverse momentum and opening angle, with the threshold values determined from the simulated samples.

The topology requirement identifies neutrino interactions with the same visible charged-particle pairs as the HNL decays, but does not use the momenta of those particles. We therefore form three observables from the two selected particles:

$$E_{\mathrm{pair}} = E_1 + E_2, \tag{1}$$

$$\vec{p}_{\mathrm{pair}} = \vec{p}_1 + \vec{p}_2, \tag{2}$$

$$p_{T,\mathrm{pair}} = \left|\vec{p}_{\mathrm{pair}} - (\vec{p}_{\mathrm{pair}} \cdot \hat{z})\hat{z}\right|, \tag{3}$$

$$\theta_{\mathrm{pair}} = \cos^{-1}\left(\frac{\vec{p}_1 \cdot \vec{p}_2}{|\vec{p}_1||\vec{p}_2|}\right). \tag{4}$$

Here E_i is the total energy, rather than the kinetic energy used for the detectability threshold. We take $\hat{z}$ to be the simulation $+z$ axis.

For a chosen mixing hypothesis, topology t, and target mass m_N, the rectangular selection is

$$E_{\mathrm{pair}} < E^t_{\mathrm{max}}(m_N), \qquad p_{T,\mathrm{pair}} < p^t_{T,\mathrm{max}}(m_N), \qquad \theta_{\mathrm{pair}} < \theta^t_{\mathrm{max}}(m_N). \tag{5}$$

All three inequalities must hold for an event to pass. The same triplet is applied to every signal parent family contributing at that mass and to the corresponding GENIE background data. The cuts are made mass dependent and optimised separately for muon and electron dominant mixing.

In order to optimise the kinematic cuts, we choose the baseline triplet by maximising the ratio of weighted signal to background efficiencies,

$$\mathcal{F}_\epsilon = \frac{\epsilon_S}{\epsilon_B}, \qquad \epsilon_S = \frac{\sum_{i \in S_{\mathrm{pass}}} w_{S,i}}{\sum_{i \in S_{\mathrm{input}}} w_{S,i}}, \qquad \epsilon_B = \frac{\sum_{j \in B_{\mathrm{pass}}} w_{B,j}}{\sum_{j \in B_{\mathrm{Stage1}}} w_{B,j}}, \tag{6}$$

subject to $\epsilon_S \geq 0.85$ and $\epsilon_B > 0$. The signal denominator contains the usable HNLux pair sample for the selected topology and target mass, and the background denominator is the complete sample for that topology. Signal weights include the stored HNLux decay probability at the reference mixing, while GENIE weights are scaled to the same 1.75×10^{19}-POT exposure used in the sensitivity calculation. The minimum signal efficiency prevents the scan from selecting an apparently strong background rejection by discarding most of the signal.

For each parent family, the scan uses an exact simulated mass where available. Otherwise its nearest lower and upper simulated masses provide brackets. Candidate thresholds are formed from weighted signal-quantile support for the parent-summed sample: 40 values per observable span quantiles from 0.85 to 0.995, before duplicate thresholds are removed. We evaluate the Cartesian product of these three grids and retain the valid triplet with the largest $\mathcal{F}_\epsilon$. Cumulative three-dimensional histograms supply the weighted pass yields and raw surviving background counts.

In Figure 4 we show a set of representative kinematic cuts, for the 2μ-like channel at $m_N = 1.018$ GeV in the muon-mixing sample, where the contributing HNL parent families are D and D_s. The jointly optimised cuts retain $\epsilon_S = 0.850$ of the weighted signal and $\epsilon_B = 0.0660$ of the Stage-1 weighted background, corresponding to 93.4% background rejection. In the histograms below we display each variable separately; the quoted efficiencies follow from applying all three cuts simultaneously.

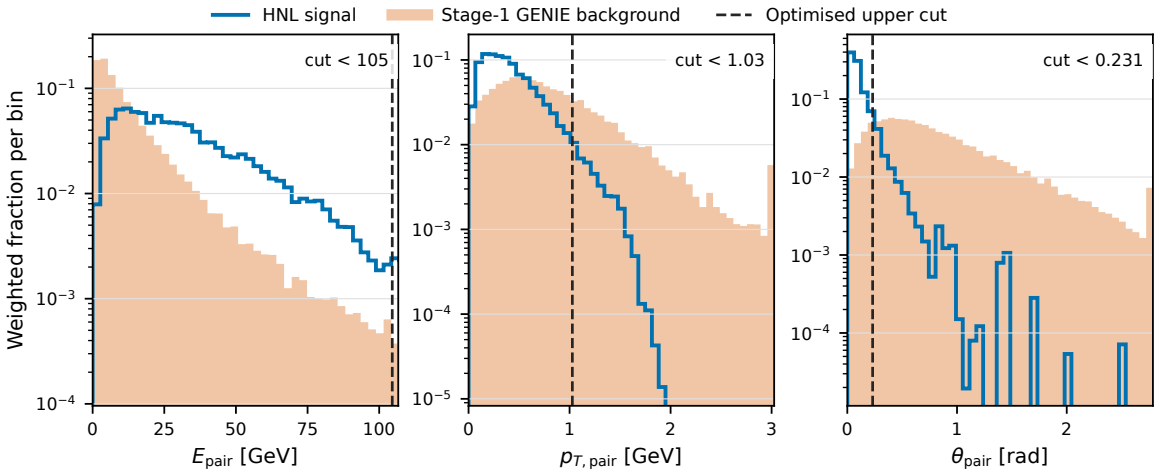


Figure 4: Weighted signal and background distributions for the 2μ-like topology at $m_N = 1.018$ GeV in the muon-mixing sample. Each distribution is normalised to unit weighted sum before the three-variable selection; the vertical lines mark the jointly optimised upper thresholds. The signal includes the D and D_s parent families. The cuts are $E_{\mathrm{pair}} < 104.6$ GeV, $p_{T,\mathrm{pair}} < 1.028$ GeV and $\theta_{\mathrm{pair}} < 0.231$ rad.

Across the simulated mass points, the selected cuts retain at least 85% of the weighted signal by construction. Signal efficiency ranges from approximately 85–99% for $2e$, and stays close to 85% for the 1μ-like $+1e$ and 2μ-like topologies. Relative to the Stage-1 weighted background, rejection ranges from approximately 83–98%, 89–99.7%, and 92–99.9%, respectively.

The ϵ_S/ϵ_B objective is useful for background rejection but does not directly minimise the expected mixing limit. While computationally expensive, further work could determine the kinematic cuts by directly optimising for a smaller projected mixing threshold for each topology. On top of this, a limitation of our method is that there may be multiple regions in the parameter space that have

distinguishing power between signal and background data. Linear cuts may not effectively capture these regions, and so further work could investigate the use of boosted decision trees in the optimisation of the kinematic cuts.

3.4 Sensitivity Calculation

So far, the analysis has determined which simulated HNL decays and neutrino interactions pass the topology and kinematic selections. To turn those selections into a sensitivity, we need the expected number of selected signal and background events. A sensitivity curve plots, against the assumed HNL mass m_N, the smallest squared mixing $|U_{\alpha N}|^2$ for which the expected selected signal reaches the nominal 90% confidence-level threshold above the selected background under the approximate rule below. A lower curve therefore represents sensitivity to a smaller mixing. At each trial mixing and mass point, we reevaluate the expected signal yield and compare it to the signal yield required for that threshold, given the expected background yield.

Signal and background yields are both normalised to a five-year benchmark exposure of $N_{\mathrm{POT}} = 1.75 \times 10^{19}$ protons on target (POT), based on the projected 3.5×10^{18} POT per year at T2 [5]. For a selected GENIE event j from incident neutrino flavour β, the background normalisation is simply

$$w_j^{\mathrm{bkg}} = w_j^{\mathrm{GENIE}} \, \frac{N_{\mathrm{int},\beta}^{\mathrm{ref}}}{N_{\beta}^{\mathrm{gen}}} \, \frac{N_{\mathrm{POT}}}{N_{\mathrm{POT}}^{\mathrm{ref}}}. \tag{7}$$

Here w_j^{GENIE} is the generator event weight, $N_{\mathrm{int},\beta}^{\mathrm{ref}}$ is the preliminary expected number of NP04 neutrino interactions of that flavour at the reference exposure [10], and $N_{\beta}^{\mathrm{gen}} = 200{,}000$ is the number of generated interactions in its sample. The last ratio scales the reference exposure $N_{\mathrm{POT}}^{\mathrm{ref}} = 1.5 \times 10^{17}$ POT to the target exposure [10]. Summing these weights over events that pass the mass-dependent cuts gives the selected background expectation $B(m_N)$.

The signal requires a production and decay weight for every HNLux event. At the generated reference mixing $|U_{\alpha N}|^2_{\mathrm{ref}} = 10^{-6}$, this is

$$w_i^{\mathrm{ref}} = P_{w,i} \, N_{w,i} \, \frac{\Gamma(P \to N)_i}{\Gamma_{P,\mathrm{norm}}} \, B_{N,i} \, \frac{N_{\mathrm{POT}}}{\mathrm{POT}_{f,i}}. \tag{8}$$

The parent-meson weight $P_{w,i}$ represents the flux of the parent reaching the HNL production stage. The factor $\Gamma(P \to N)_i / \Gamma_{P,\mathrm{norm}}$ converts this flux weight to the HNL-production contribution using the normalisation of the parent sample: $\Gamma_{P,\mathrm{norm}}$ is the inclusive light-neutrino partial width for pion and kaon parents, and the total parent width for D and D_s parents. The HNL branching fraction $B_{N,i}$ selects the visible decay mode, while $N_{\mathrm{POT}}/\mathrm{POT}_{f,i}$ scales the HNLux file exposure to the analysis exposure. Finally, $N_{w,i}$ is the HNLux probability for the HNL to decay in the detector at the reference mixing [8, 7]. These reference weights are used when choosing the cuts; the final sensitivity scan recalculates the decay probability for each trial mixing.

For an HNL travelling towards NP04, the decay probability used in the scan follows the standard exponential survival and decay law [5, 4]:

$$P_{\mathrm{dec},i}(|U_{\alpha N}|^2) = e^{-L_i / L_{\mathrm{lab},i}(|U_{\alpha N}|^2)} \left[1 - e^{-\Delta \ell_i / L_{\mathrm{lab},i}(|U_{\alpha N}|^2)} \right]. \tag{9}$$

The first factor is the probability of surviving the distance L_i from production to detector entry; the bracket is the probability of decaying over the in-volume interval $\Delta \ell_i$. In this calculation, that interval

ends at the corrected sampled HNLux decay vertex: we subtract one NP04 centre displacement from the stored wNP04 coordinates to place the vertex in the active-volume box, then find detector entry along the ray from the production point. The lab-frame decay length is $L_{\mathrm{lab},i} = \beta_i \gamma_i c \tau_N$: it combines the HNL lifetime with its boost. Under the single-flavour mixing hypothesis, the production partial width and total HNL width both scale with $|U_{\alpha N}|^2$, while the decay branching fraction stays fixed [7]. Consequently, with $r_\alpha = |U_{\alpha N}|^2 / |U_{\alpha N}|^2_{\mathrm{ref}}$, the trial weight is

$$w_i(|U_{\alpha N}|^2) = P_{w,i}\, \frac{\Gamma(P \to N)_i^{\mathrm{ref}}}{\Gamma_{P,\mathrm{norm}}}\, B_{N,i}\, \frac{N_{\mathrm{POT}}}{\mathrm{POT}_{f,i}}\, r_\alpha\, P_{\mathrm{dec},i}(|U_{\alpha N}|^2), \qquad L_{\mathrm{lab},i}(|U_{\alpha N}|^2) = \frac{L_{\mathrm{lab},i}^{\mathrm{ref}}}{r_\alpha}. \tag{10}$$

Thus increasing the mixing increases HNL production but also shortens its lifetime; at sufficiently large mixing, more HNLs decay before reaching NP04 [3]. For each mass and trial mixing we sum the weights of events passing the cuts within each parent family, interpolate the family yields between simulated masses, and then sum the families to obtain $S(|U_{\alpha N}|^2, m_N)$.

We use a background-only expected observation and the following approximate one-bin requirement for a nominal 90% confidence-level reach:

$$S_{\mathrm{req}}(B) = \max\left[2.44,\ \sqrt{4.61\,(B + (0.2B)^2)}\right]. \tag{11}$$

Here B is the selected background yield, B inside the square root is its counting variance, and $(0.2B)^2$ represents an assumed 20% background-normalisation uncertainty, taking the size of the flux-normalisation uncertainty used in the ICARUS-like HNL study as a benchmark [4]. The factor 4.61 uses that study's two-parameter 90% χ^2 convention [4], while 2.44 supplies the zero-background Feldman–Cousins counting floor [14]. Their combination in the maximum above is our approximate one-bin prescription, rather than a likelihood calculation with demonstrated 90% coverage for NP04. At each mass, the plotted reach is the first crossing of $S(|U_{\alpha N}|^2, m_N) = S_{\mathrm{req}}[B(m_N)]$ as the mixing increases.

4 Results

In this section we present the projected sensitivity of each visible topology under muon- and electron dominant mixing and discuss the structure and limitations of the curves. The selected signal and GENIE background yields correspond to a normalisation of 1.75×10^{19} POT.

Figures 5–7 show the nominal 90% confidence-level sensitivity curves for the three two-particle topologies, with muon mixing on the left and electron mixing on the right of each figure. In each plot, the coloured solid curve is the signal-only projection for that topology, while the coloured dashed curve includes the selected GENIE neutrino background and a 20% normalisation uncertainty. The black curve shows a separate signal-only reference using six simulated decay channels, including modes outside the three analysis topologies. The process described in 3.4 is used to calculate the expected yield and sensitivity at each mass point with data within the plotted range; yields are then grouped by HNL-parent family and interpolated between the simulated masses. The numerical values discussed below are recorded in the analysis output tables.

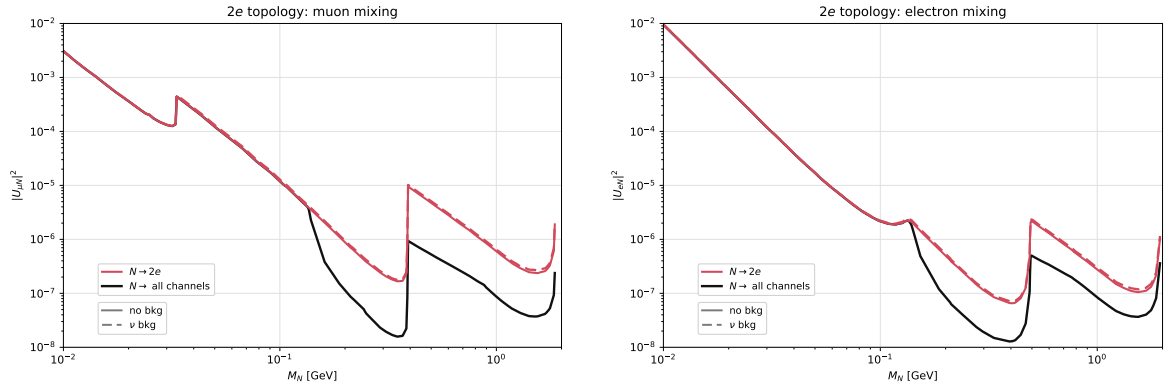


Figure 5: Expected $|U_{\mu N}|^2$ and $|U_{eN}|^2$ sensitivities respectively for the $2e$ topology at 1.75×10^{19} protons on target. The red solid curve is the signal-only result for this topology; the red dashed curve includes selected GENIE background events.

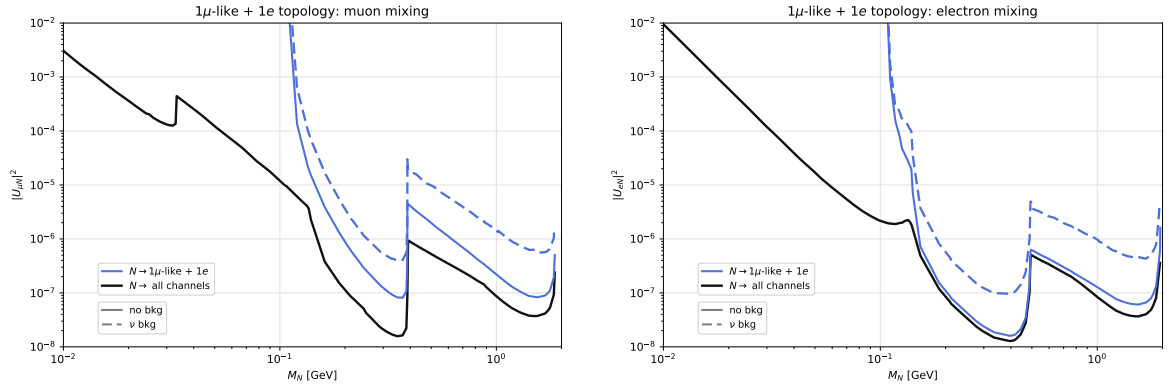


Figure 6: Expected $|U_{\mu N}|^2$ and $|U_{eN}|^2$ sensitivities respectively for the 1μ-like $+1e$ topology. The blue solid curve is the signal-only result for this topology; the blue dashed curve includes selected GENIE background events.

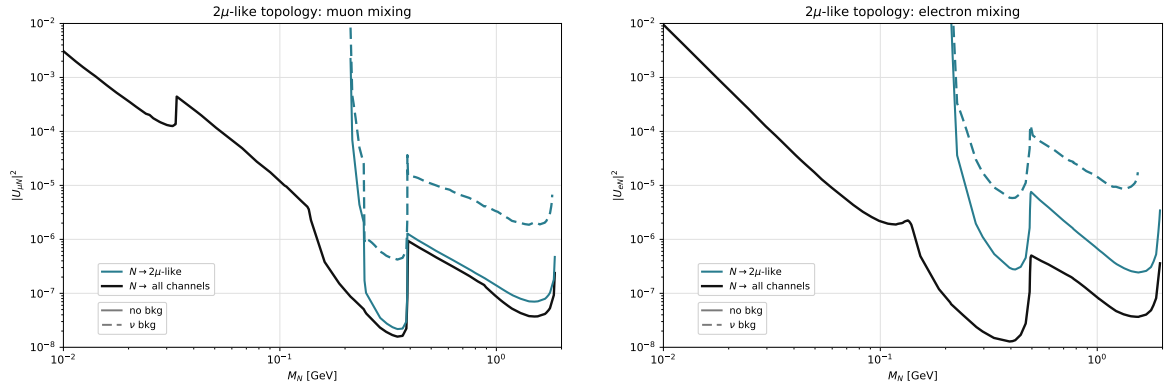


Figure 7: Expected $|U_{\mu N}|^2$ and $|U_{eN}|^2$ sensitivities respectively for the 2μ-like topology. The teal solid curve is the signal-only result for this topology; the teal dashed curve includes selected GENIE background events.

For muon mixing, the $2e$ topology gives the strongest muon-mixing reach over most of the covered mass range. Its lowest plotted background-aware value is $|U_{\mu N}|^2 = 1.8 \times 10^{-7}$ at $m_N = 0.349$ GeV. The mixed and 2μ-like topologies reach approximately 4.0×10^{-7} and 4.2×10^{-7}, respectively, near the same mass; the 2μ-like topology begins only above its visible-channel threshold.

The abrupt weakening near $m_N = 0.39$ GeV coincides with the end of the kaon-parent contribution. In the $2e$ sample, the selected signal expectation at the reference mixing falls from 2.76 events at 0.388 GeV, where K, D and D_s contribute, to 0.0234 events at 0.391 GeV, where only D and D_s remain. The corresponding selected background changes much less, from 0.376 to 0.222 expected events. The loss of kaon-supported signal therefore explains most of this sharp feature.

For electron mixing, the $2e$ topology reaches the lowest background-aware value overall, $|U_{eN}|^2 = 7.0 \times 10^{-8}$ at $m_N = 0.406$ GeV. It also covers the lowest simulated masses through $N \to \nu e^+ e^-$. The 1μ-like $+1e$ topology becomes available above the electron–muon decay threshold and reaches 9.6×10^{-8} at 0.394 GeV. At some masses between approximately 0.21 and 0.30 GeV, its projected reach is marginally stronger than that of $2e$, although $2e$ gives the stronger minimum overall. The 2μ-like topology begins above the dimuon threshold and, in the electron-mixing samples, contains only $N \to \nu\mu^+\mu^-$; its best projected reach is 5.8×10^{-6} at 0.406 GeV.

The electron-mixing curves weaken near 0.49–0.50 GeV as the kaon-parent contribution ends, at a higher mass than in the muon-mixing case. For $2e$, the selected signal expectation at the reference mixing falls from 9.94 events at 0.488 GeV, with K, D and D_s parents, to 0.466 events at 0.504 GeV, with only D and D_s parents. The corresponding selected background changes from 0.760 to 0.350 expected events. The remaining charm-parent contributions improve the $2e$ reach again over much of the higher-mass region before it weakens near the upper end of the simulated range.

The dashed $2e$ curve nearly follows its signal-only counterpart, while the separation is larger for the mixed and 2μ-like topologies. That separation includes both the selected background and the change in signal acceptance caused by the kinematic cuts, so a background-only penalty requires a topology-matched comparison.

5 Conclusion

The purpose of this report was to evaluate the expected sensitivity of ProtoDUNE-HD to heavy neutral leptons (HNLs) using simulated HNL decays and beam-induced neutrino backgrounds. We considered three two-charged-particle topologies, $2e$, 1μ-like $+1e$ and 2μ-like. For the neutrino background we required exactly two detectable charged particles, then used mass-dependent pair kinematics to distinguish those events from the HNL signal. We calculated the expected signal and background yields at 1.75×10^{19} protons on target, accounting for HNL production by the simulated parent mesons and the probability of decay inside NP04. This gave us a common basis for comparing electron- and muon-mixing sensitivities.

The $2e$ topology provided the strongest expected reach in both benchmarks: $|U_{eN}|^2 = 7.0 \times 10^{-8}$ at $m_N = 0.406$ GeV for electron mixing and $|U_{\mu N}|^2 = 1.8 \times 10^{-7}$ at $m_N = 0.349$ GeV for muon mixing. The 1μ-like $+1e$ and 2μ-like topologies probe other visible decays but have weaker overall minima. The curves also show the importance of the HNL production channel: in the muon-mixing case, sensitivity weakens sharply near $m_N = 0.39$ GeV when the kaon-parent contribution disappears. These results show how the available signal channels and selected neutrino background shape the expected reach at different masses.

Before a final experimental reach can be claimed, the analysis must include detector response, reconstruction and particle identification, cosmic-ray backgrounds, an independently validated neutrino-background estimate and a combined statistical treatment of the three topologies. These steps will establish what a realised ProtoDUNE exposure can contribute to the search for long-lived neutral particles.

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Research and Writing Methods

Generative AI tooling, primarily LLM model "ChatGPT-5.6 sol", was used for grammatical and sentence structure improvements in the main body of the report, as well as LaTeX formatting improvement tasks. Otherwise, all analysis, figures and text in the report were produced by the author, unless explicitly stated otherwise.

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